TRAC 2026 · DIGITAL EPIGRAPHY

No Longer Graveyards for Historians but still Treacherous Terrain

A computational test of Roman marriage-age variation, 50–349 CE

Adela Sobotkova
Aarhus University · adela@cas.au.dk

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Ages in inscriptions biased — a “graveyard for historians”

> *“Roman tombstones are intractable evidence — the ages they record cannot be taken at face value.” (Hopkins 1987, paraphrased)*

The problem

Epigraphic ages are systematically distorted by who gets commemorated and how — not by who actually died at those ages.

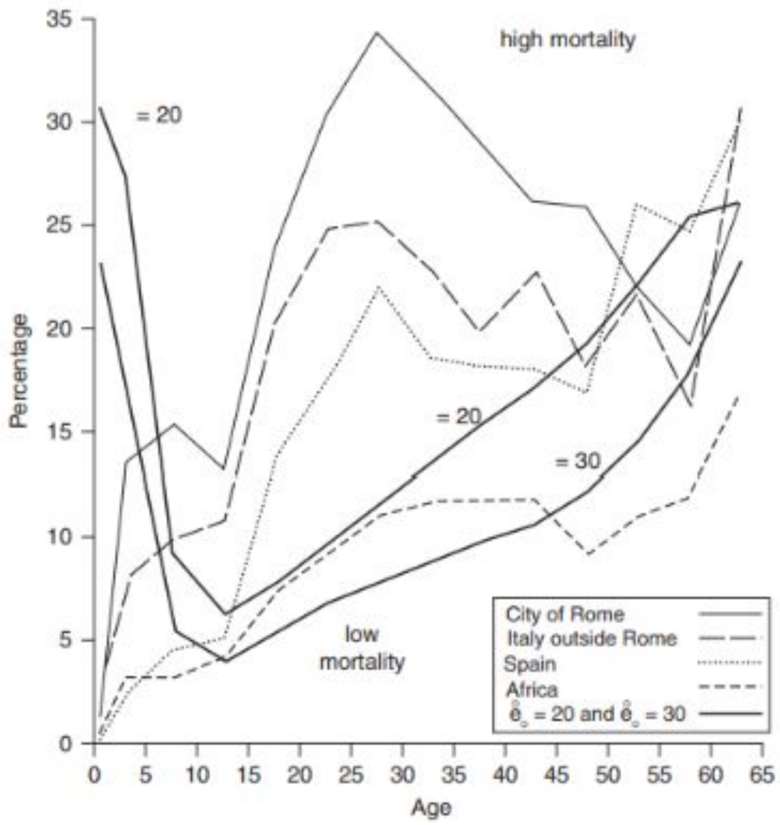


Figure 4.3. Graph showing females dying in successive age groups as percentage of those surviving to the beginning of that age group in two UN model life tables ($\mathring{e}_o$ = 20 and 30) and from inscriptions from four areas of the Roman empire.

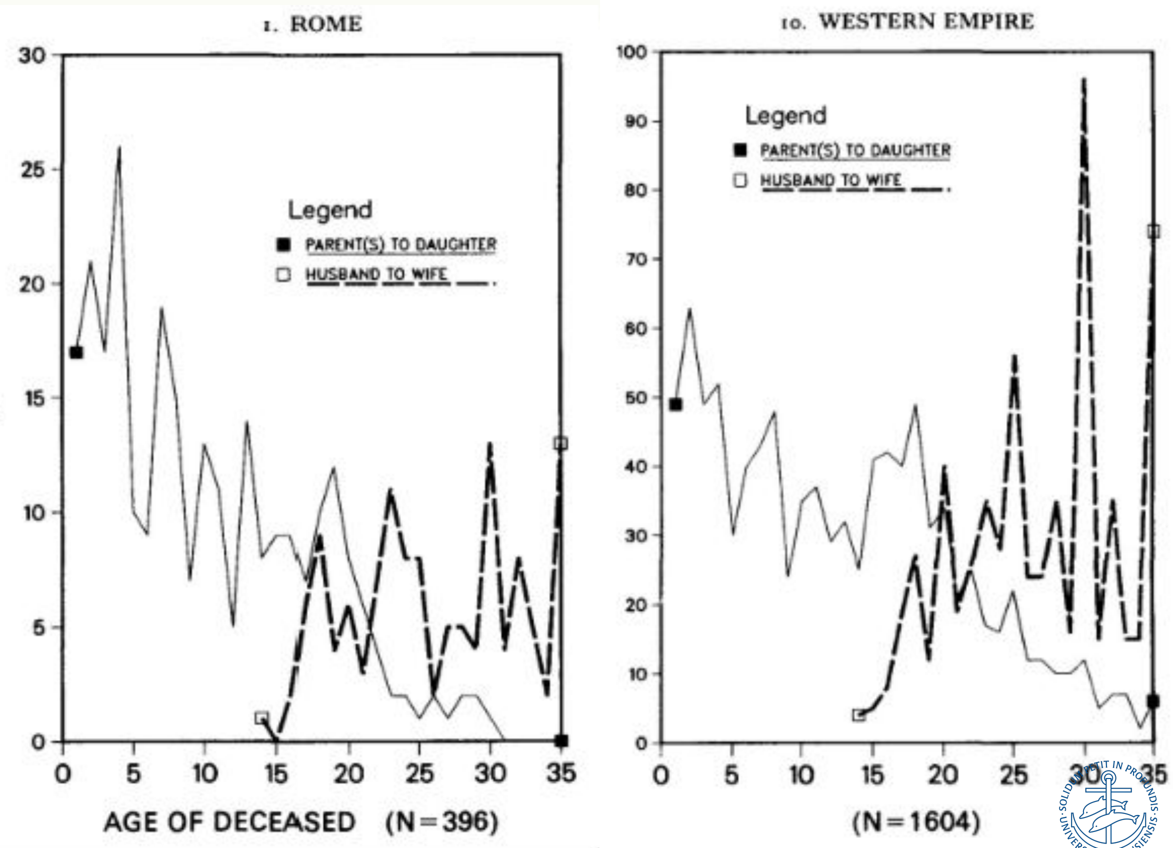
Hopkins, Keith. 1987. “Graveyards for Historians.” In F. Hinard, *La Mort, Les Morts et L’au-Delà Dans Le Monde Romain. Actes Du Colloque de Caen, 20–22 Novembre 1985*, 113–26. Caen: Centre de publications de l’Université de Caen.

Shaw and Saller didn't fix the bias — they exploited it

Opiniae Marci filiae Neptillae annorum XIIII virgini desperatae prope diem nuptiarum defunctae Marcus Opinius Rufus et Gellia Neptilla parentes

The commemorative shift

Girls were commemorated by their parents before marriage, by their husbands after. The age at which **wife** dedications overtake **daughter** dedications is a proxy for first marriage.



Saller, Richard P., and Brent D. Shaw. 1984. "Tombstones and Roman Family Relations in the Principate: Civilians, Soldiers and Slaves." *The Journal of Roman Studies* 74 (November):124–56

Shaw, Brent D. 1987. "The Age of Roman Girls at Marriage: Some Reconsiderations." *The Journal of Roman Studies* 77 (November):28–46.

Could the crossover have moved in time/across urban-rural gap?

> *"Marriage age may have shifted from century to century, responding to disease environment and economic opportunity …" (Scheidel 2007, 400)*

B. Shaw asked ***where*** the crossover sits. Forty years on, it's time to check ***whether it moved*** — and whether its movement tracks the empire's economic and epidemiological history.

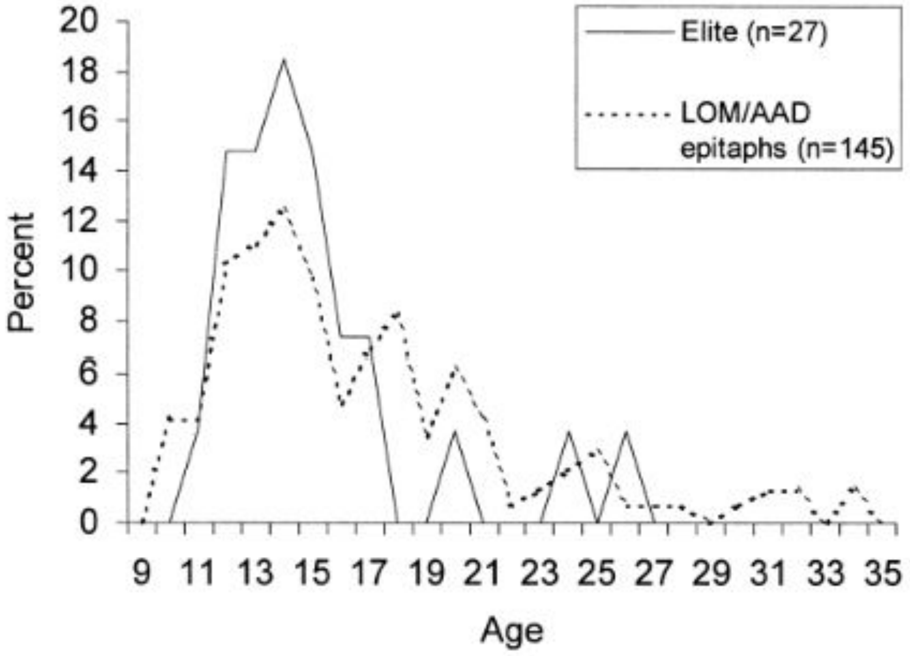


Fig. 1.—Female age of marriage in the Roman leadership and according to LOM/AAD epitaphs (Lelis et al., *Age of Marriage*, 122; Hopkins, "Age of Roman Girls," 321).

Scheidel, Walter. 2007. "Roman Funerary Commemoration and the Age at First Marriage." *Classical Philology* 102 (4): 389–402. https://doi.org/10.1086/588506.

BAUMARD 2019; CHISHOLM 1993; NETTLE 2010; WRIGLEY & SCHOFIELD

Life History Theory and Marriage Patterns as theoretical frame

Harshness

low resource availability (mean)

Unpredictability

mortality shocks, instability (variance)

Developmental calibration

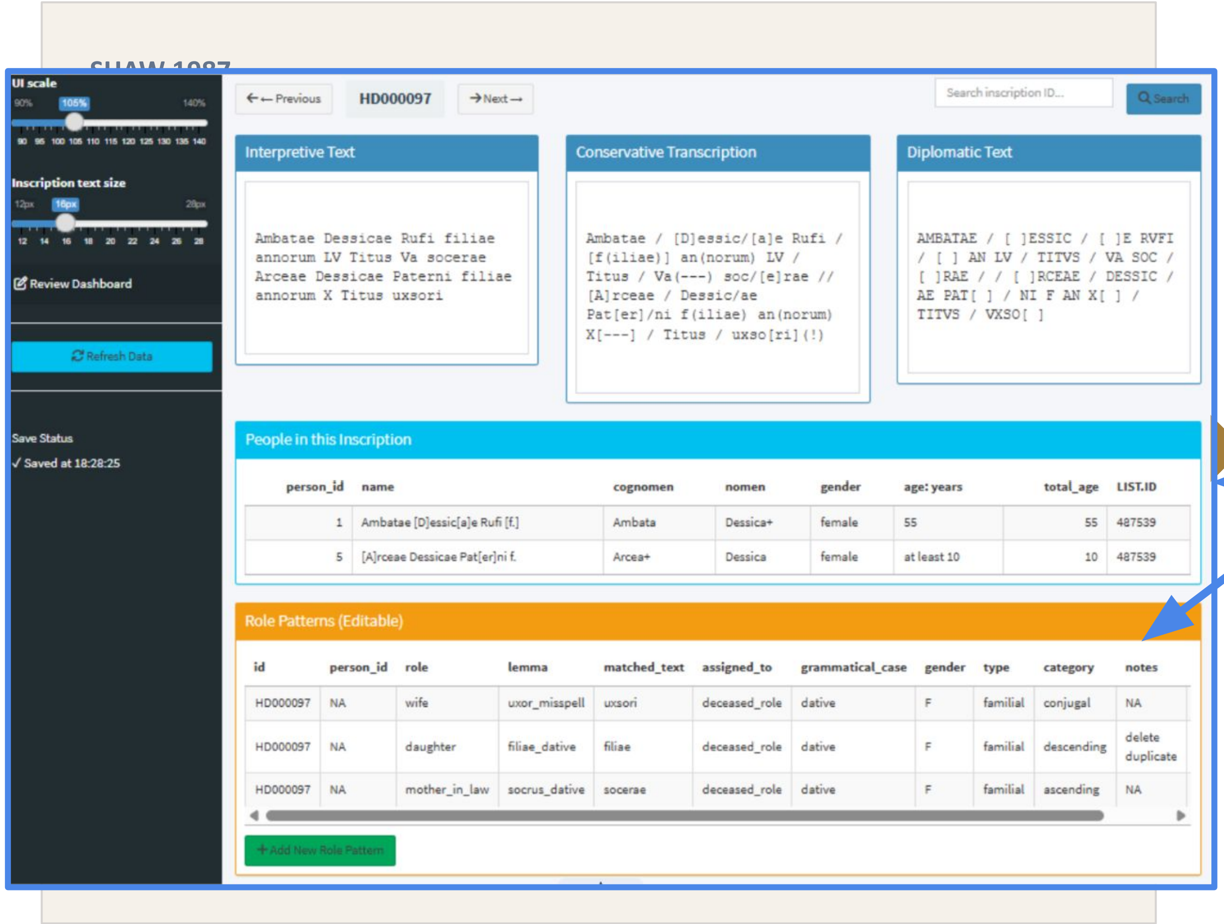
(Chisholm 1993, Nettle 2010)

Faster life history

Earlier marriage ↓ crossover age

Hypotheses 1 & 2. Pandemic shocks (Antonine, Cyprian) and economic contraction in the C2 & C3 should accelerate downward shifts in crossover age — even before absolute resources collapse. LHT predicts dropping marriage age as the Empire declines. The drop will be faster in the cities.

Why this question is newly tractable

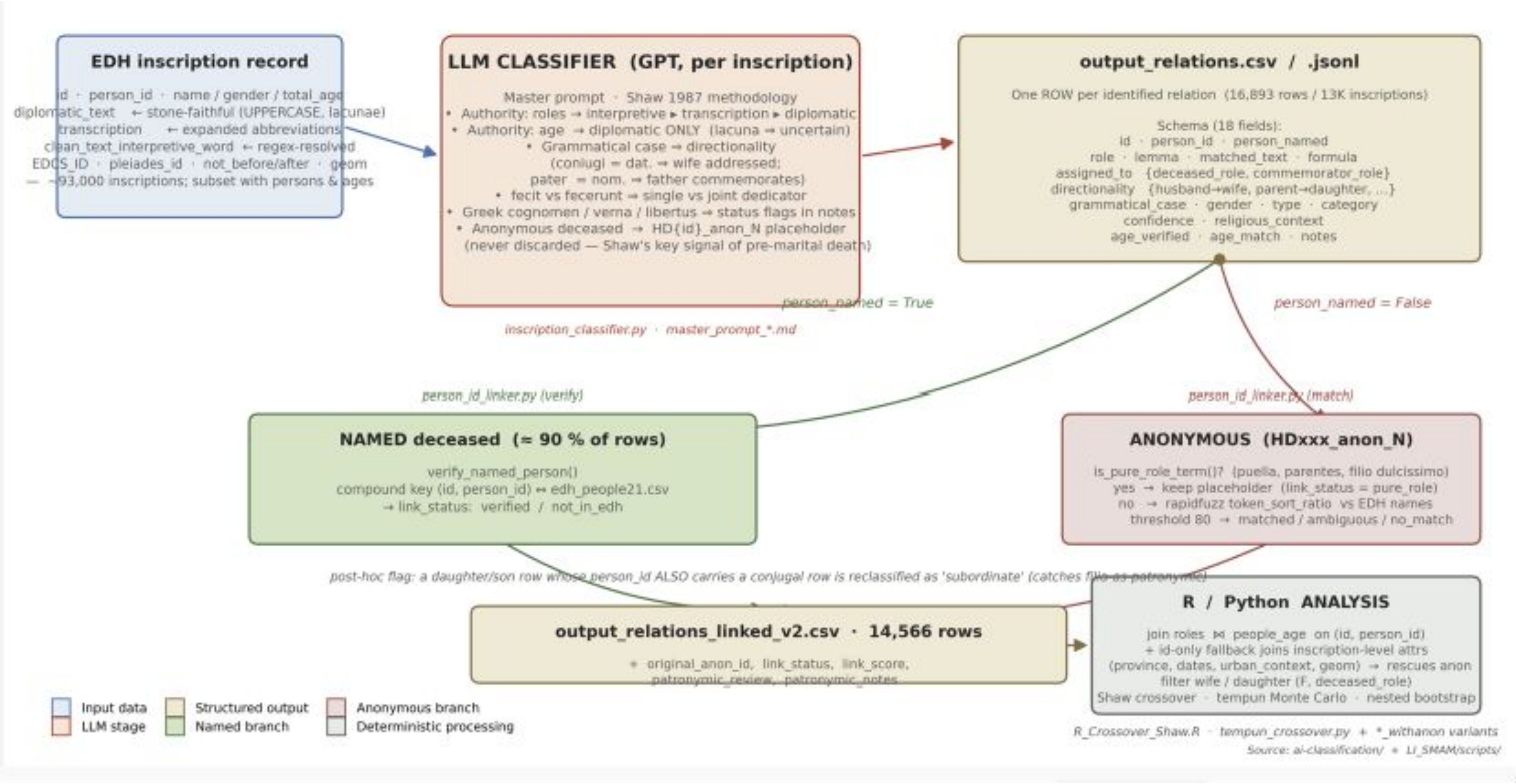


THIS PAPER

Computational pipeline

- EDH people + LIRE attributes (digital, streamlined, georeferenced)
- Encoded with structured family-relationships (human + LLM)
- Settlement-size drives spatial clustering/sampling
- tempun Monte Carlo for date uncertainty
- Auditable, longitudinal

LLM family-role classification pipeline: Gemini 3.0 + Opus 4.7



Filia patronymic and age need (human!) verification

HD000952 · id=HD000952 · person_id=2 · role=*daughter* · lemma=*filia* · case=nominative · total_age=**19** · age_match=uncertain

MATCHED TEXT

Herennia Luci filia Tertia

INSCRIPTION (CLEAN / INTERPRETIVE)

> *Caius Luxsilius Cai filius Pomptina Macer arma haec quae cernis princeps ludendo fui id ita fuisse campus urbis testis est vixit annis XVIIII Herennia Luci filia Tertia talarum vitilem ut illi cordi forem vixit annis II*

HD021926 · id=HD021926 · person_id=1 · role=*daughter* · lemma=*filia* · case=nominative · total_age=**50** · age_match=match

MATCHED TEXT

Adnamata Carveicionis filia

INSCRIPTION (CLEAN / INTERPRETIVE)

> *Adnamata Carveicionis filia annorum L hic sita est Claudius Cesoris coniugi piissimae titulum memoriae posuit*

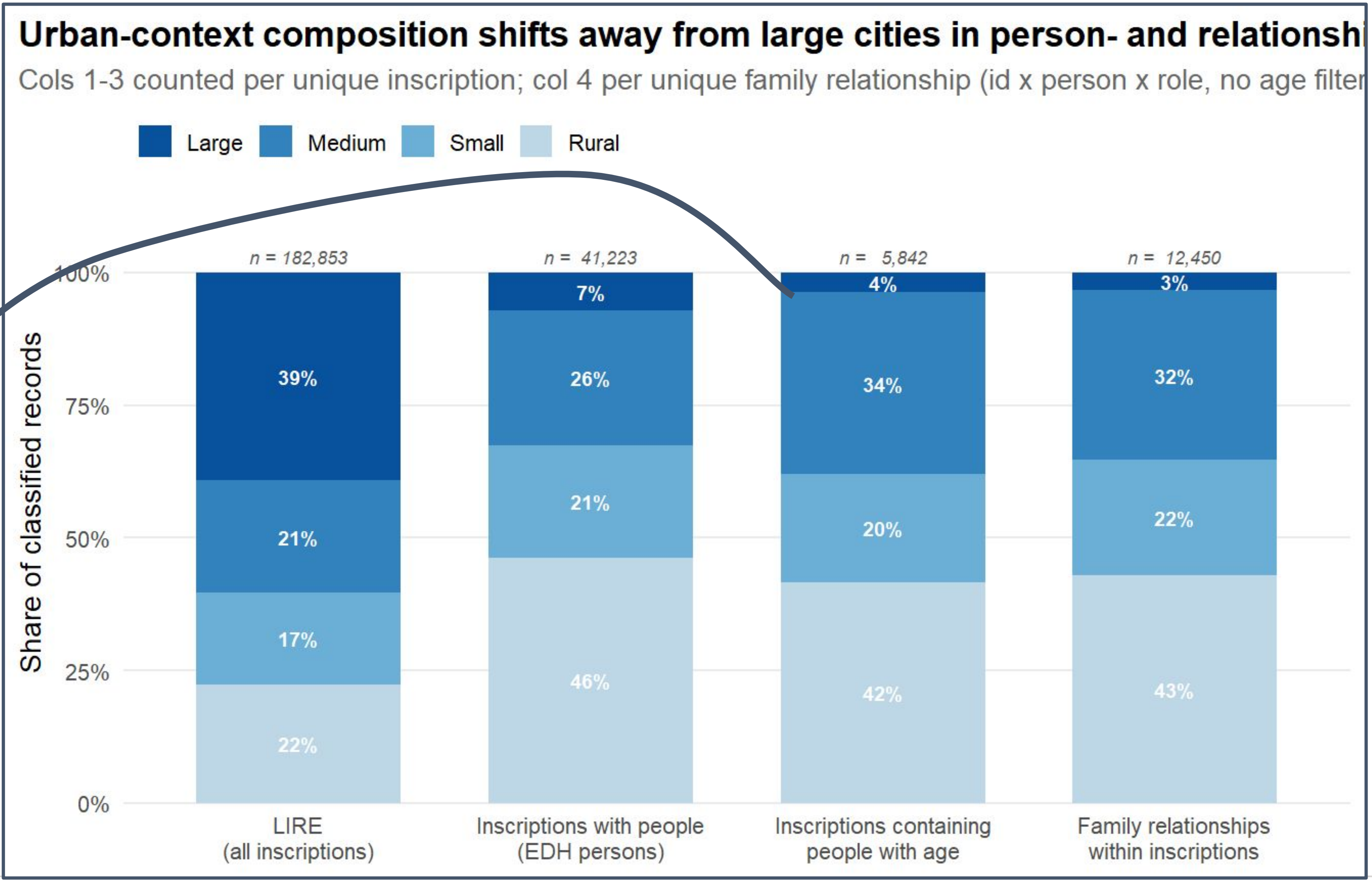
DIPLOMATIC

> ADNAMATA / CARVEICIONI / / S / / F ANN L H S E / CL CESORIS / COIVGI P / T M P

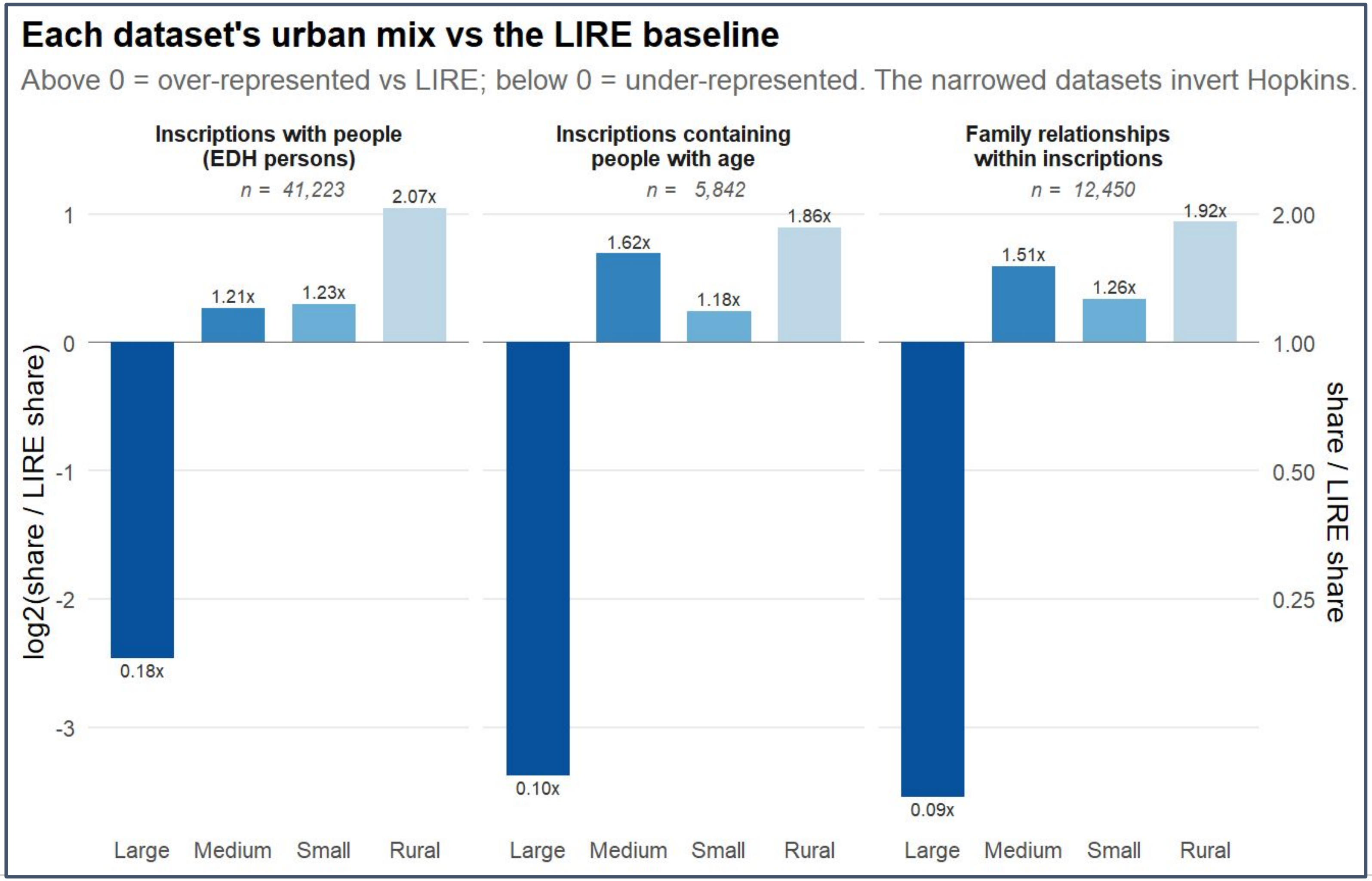
Inscriptions with people's ages and family roles not so urban

Conjugal/Descending relations in inscriptions spread across 717 unique cities/places:

large	13
medium	68
small	156
rural	480



Difference in spatial distribution of data with people from LIRE

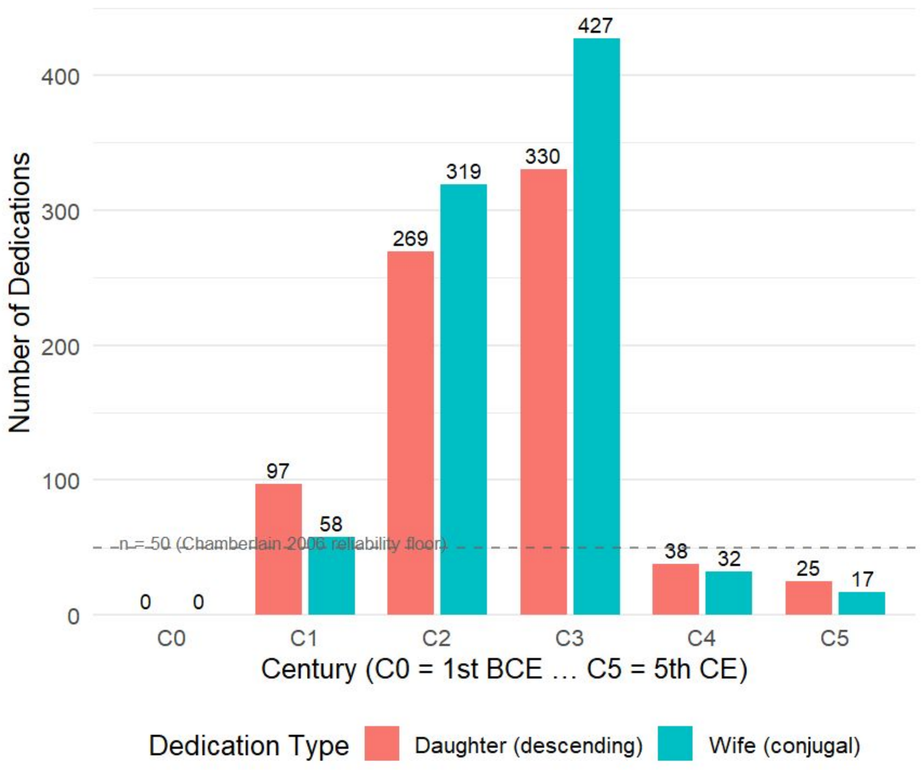


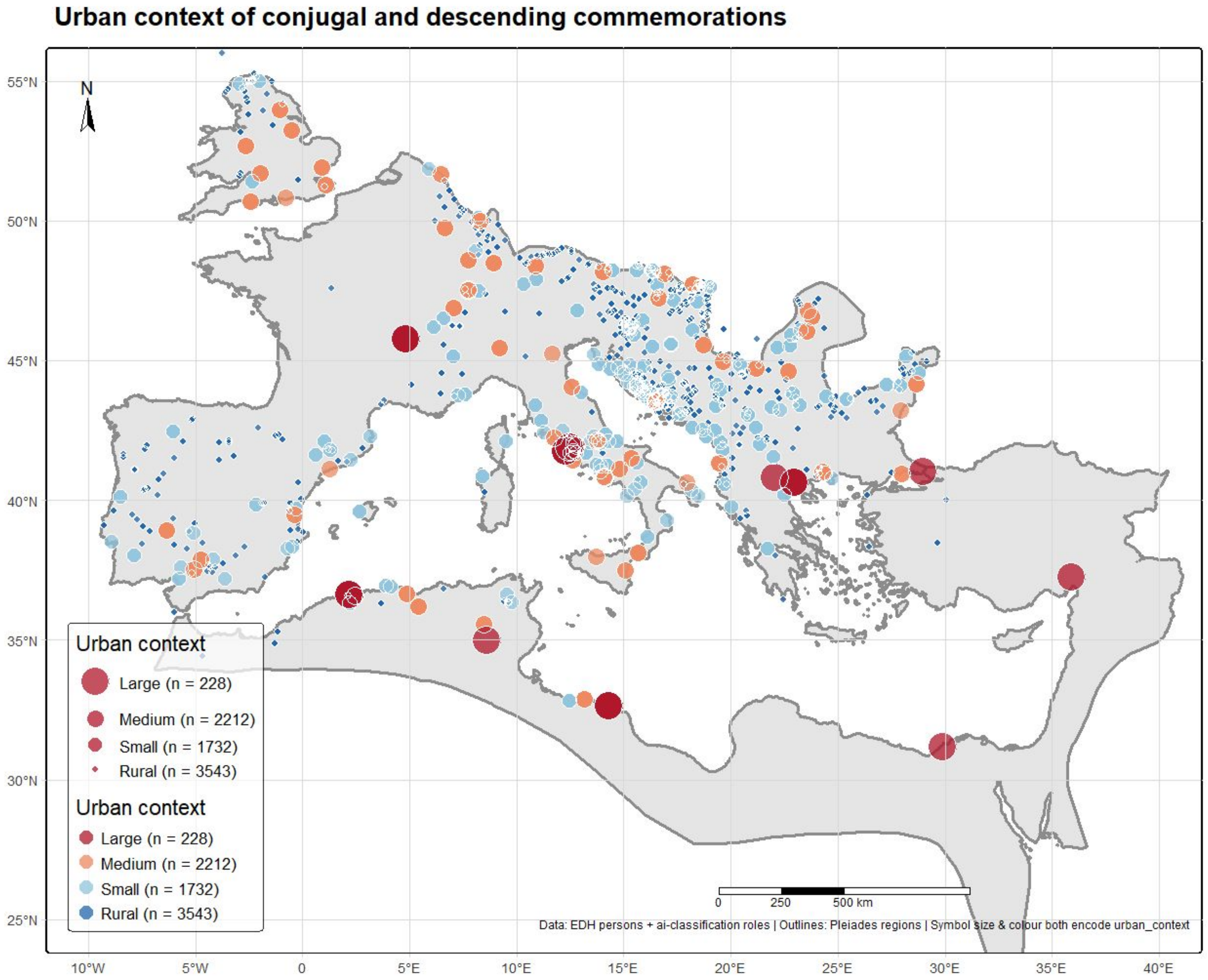
The data: wives and daughters in time and space

Deceased women named with age: 1707

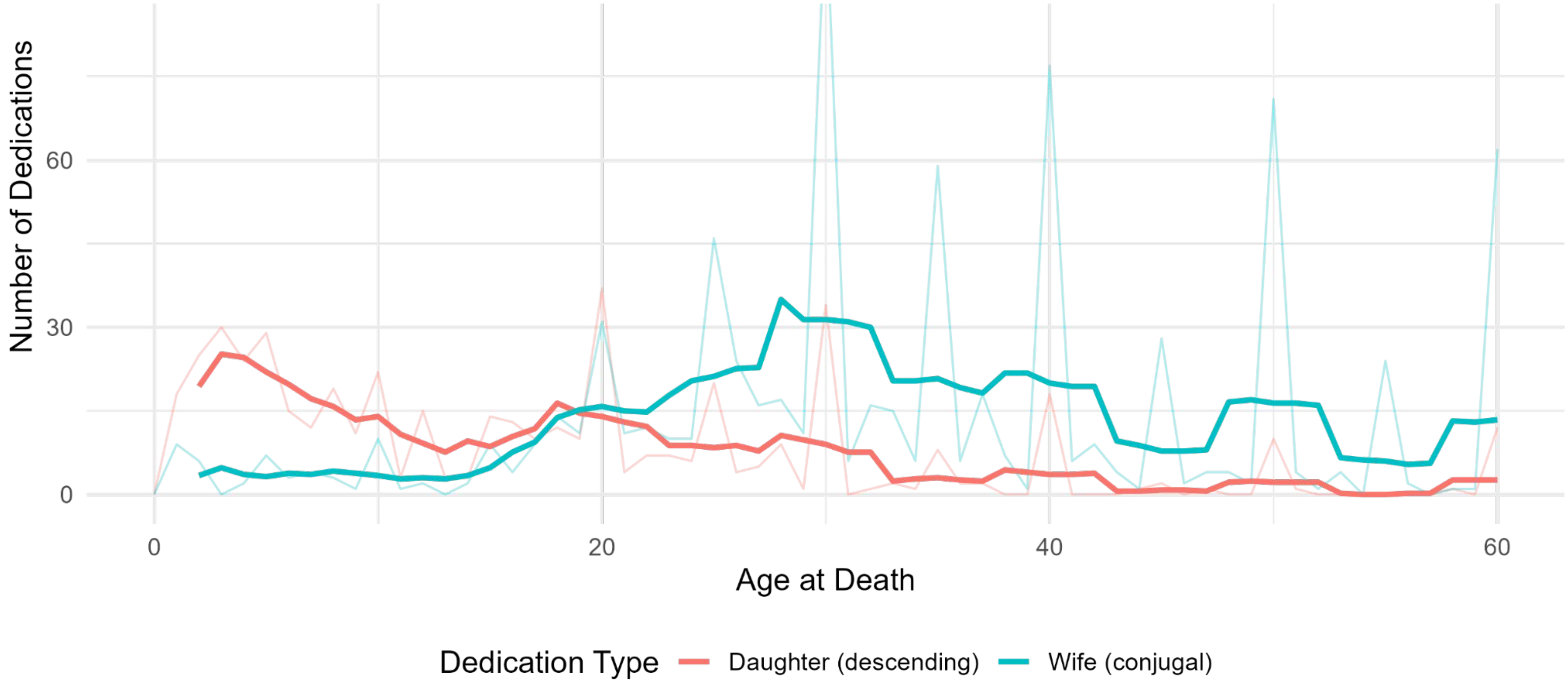
Wives: 894

Daughters: 813





Reproducing Shaw's crossover as one static point

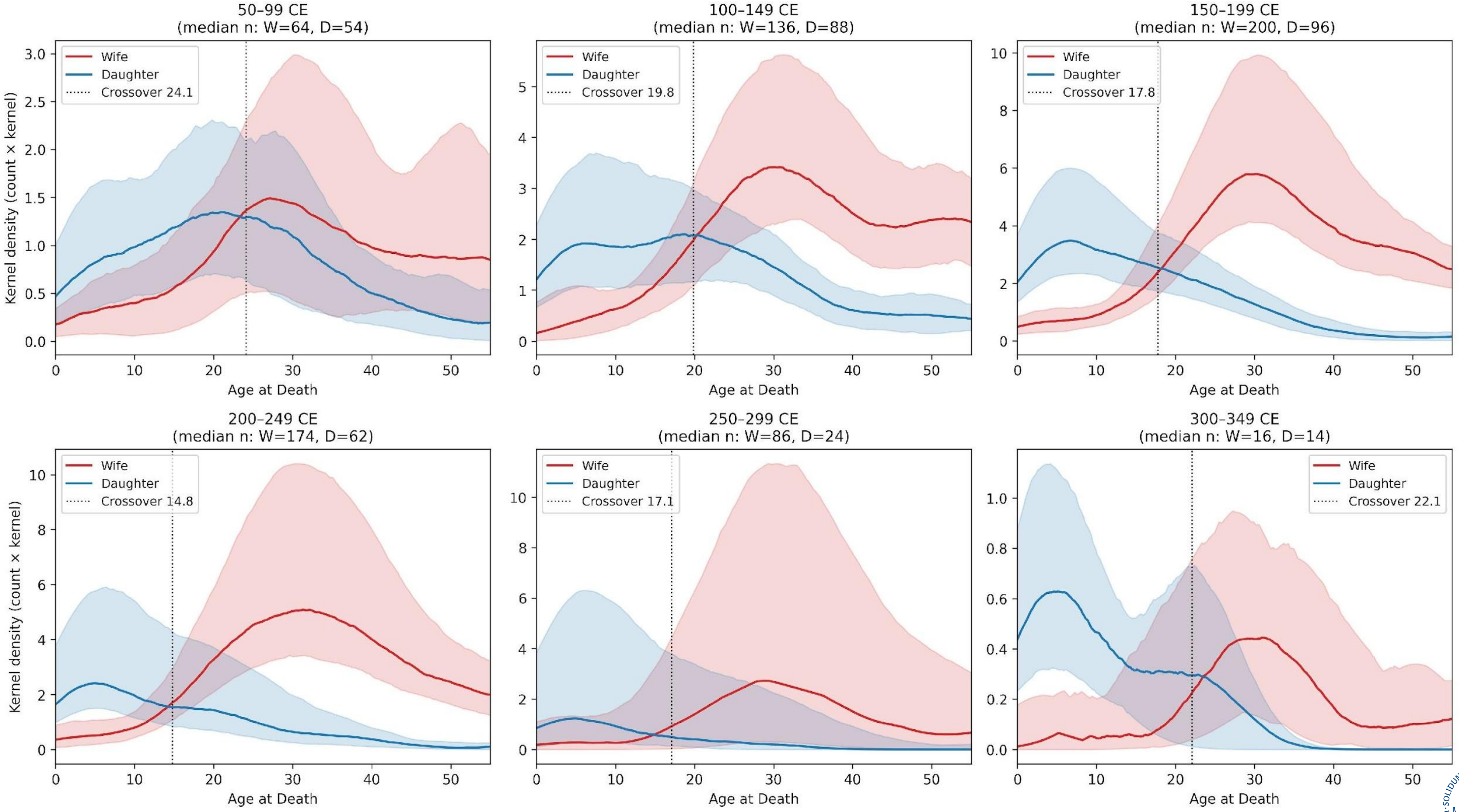


Despite much stricter spatio-temporal filters, Shaw's results reproduce.

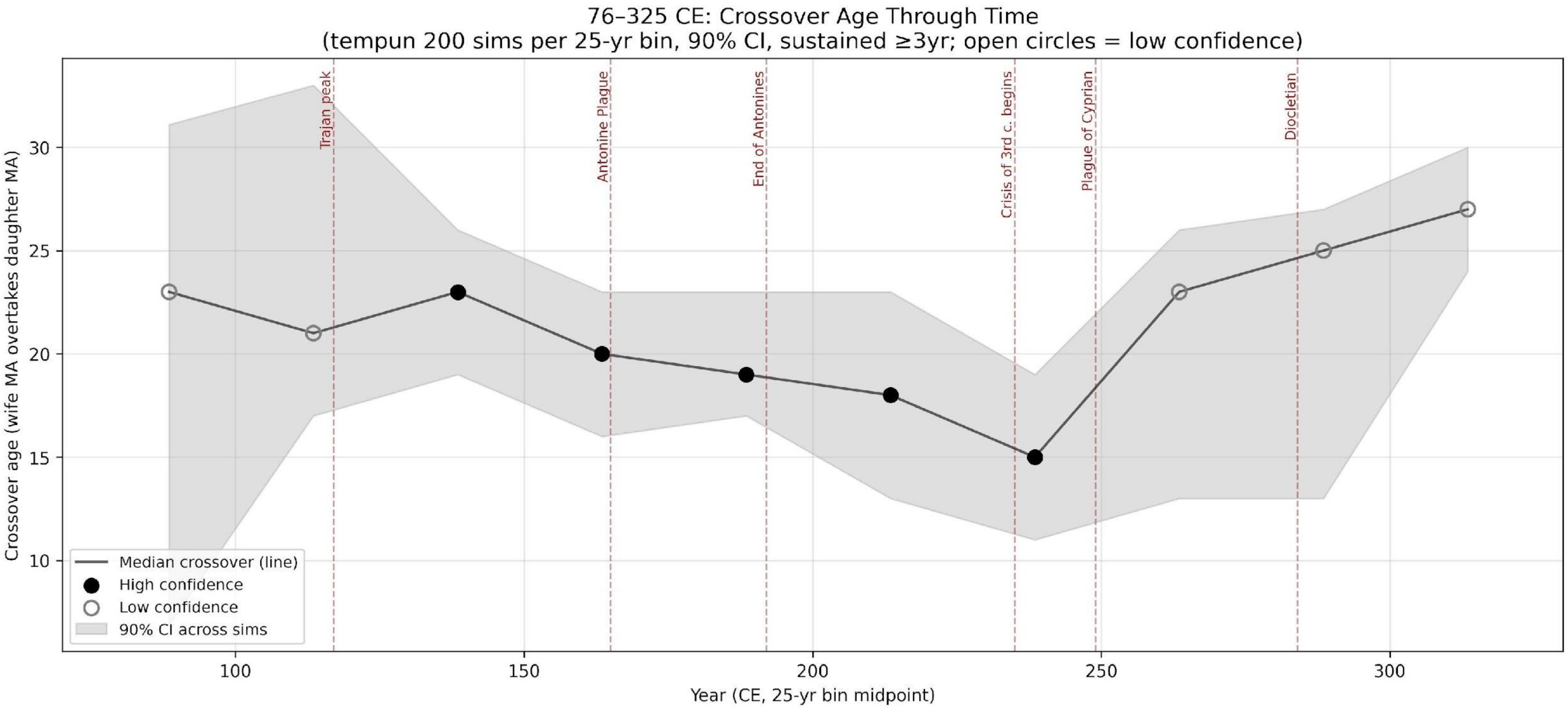
Time: crossover age in 50 years intervals

Crossover age gradually drops from ~23 years in the high imperial era to ~15 by the middle-third century AD.

These are proxies not exact years, but they are dropping, nonetheless..

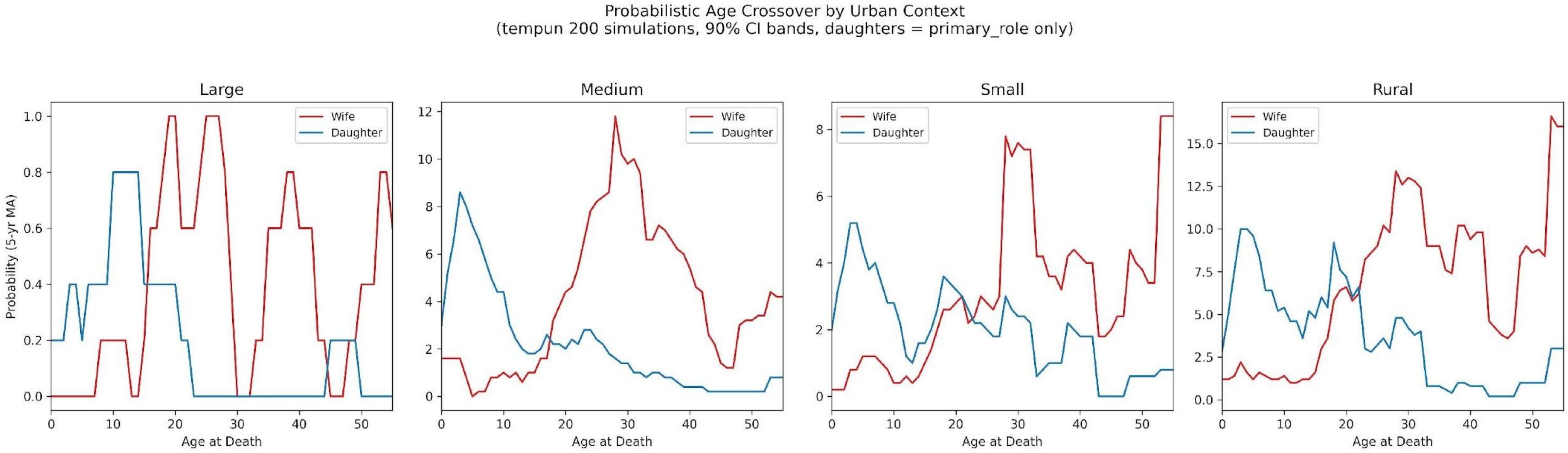


Time: Crossover age trend over centuries



Crossover age drops from ~23 in the late C1 to ~15 in the mid-C3 — the deepest trough sits between the Antonine Plague and the Plague of Cyprian.

SPACE: Crossover age is ***inverse*** to urban context size



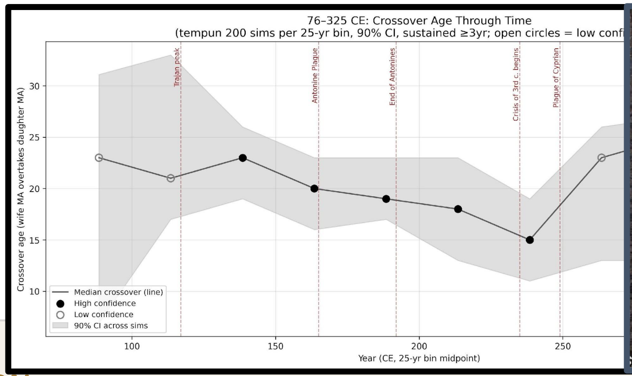
Large cities
Earliest crossover
(small n — read cautiously)

Medium / Small
Clean crossover near the aggregate (~19–20)

Rural
Latest crossover; daughter peak holds longer

Consistent with LHT (urban harshness → faster strategy) and with elite alliance dynamics (?) in cities.

Alignment with empire-scale conditions



ECONOMIC

Resource availability

C1 - early C2. Growth, peak Trajanic prosperity (surplus concentrated within the 'fortunate decile' - the top 10%)
Late C2 onward. Fiscal pressure, declining welfare ratios, 'age of anxiety', 'limits of empire'.
C3. Currency collapse, contraction.

Scheidel, Walter, and Steven J. Friesen. 2009. "The Size of the Economy and the Distribution of Income in the Roman Empire*." *The Journal of Roman Studies* 99 (November): 61–91.

EPIDEMIOLOGICAL

Mortality shocks (K. Harper)

165 CE. Antonine Plague

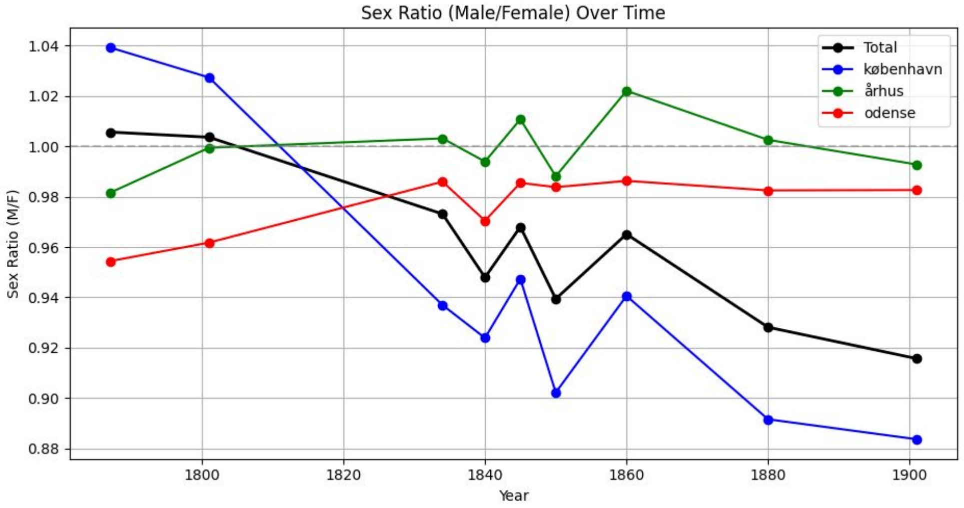
235–284 CE. Crisis of the Third Century

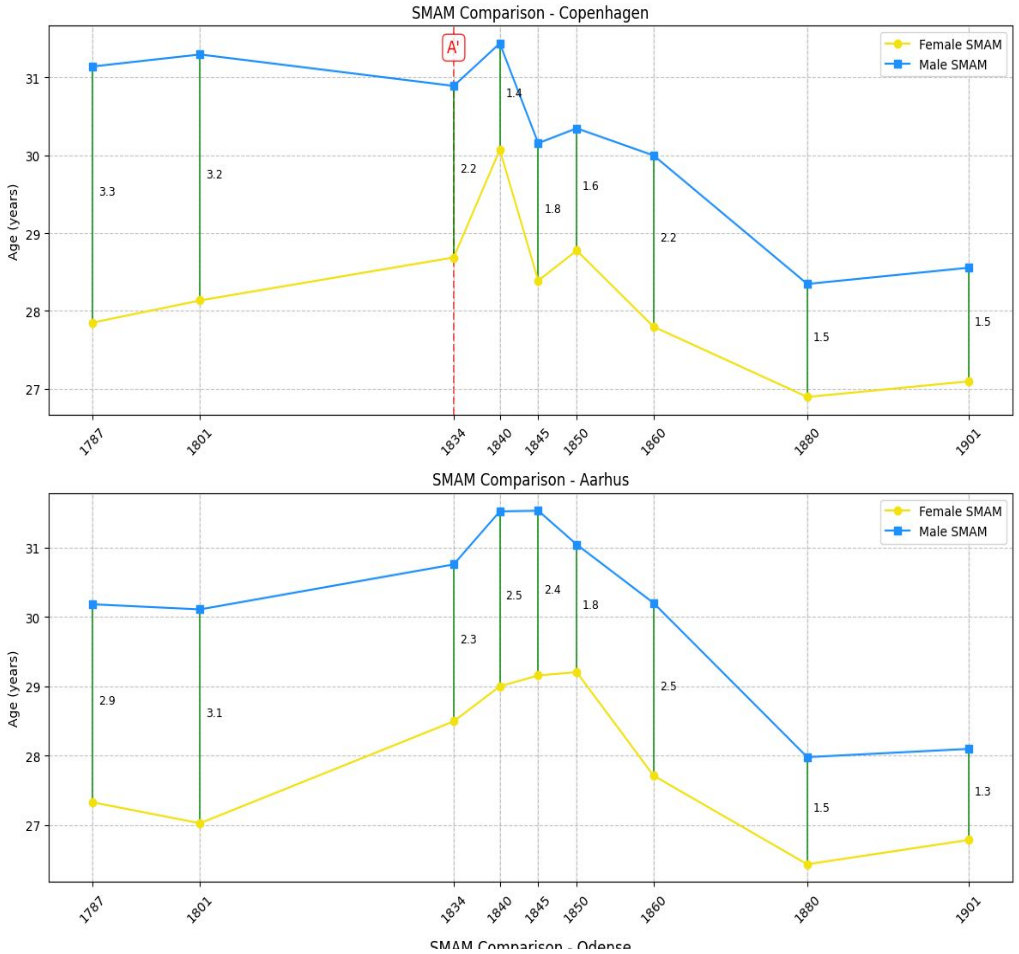
249–262 CE. Plague of Cyprian.

Harper, Kyle. 2017. *The Fate of Rome: Climate, Disease, and the End of an Empire*. Princeton University Press.

WRIGLEY & SCHOFIELD 1981; WRIGLEY, DAVIES, OEPPEN, SCHOFIELD 1997; DE MOOR and VAN ZANDEN 2010; DENNISON & OGILVIE 2014; BAUMARD 2019

Marriage patterns and economy: Not a neat correlation!





19th century Danish population data (credit to M.Nielsen)

E. A. Wrigley and Roger Schofield 1981 *The Population History of England, 1541–1871;* Wrigley, Davies, Oeppen, and Schofield, 1997 *English Population History from Family Reconstitution, 1580–1837.*
de Moor T. and J.L. van Zanden 2010: "Girl Power: The European Marriage Pattern and Labour Markets in the North Sea Region in the Late Medieval and Early Modern Period", *Economic History Review* 63 (1): 1–33
Dennison, Tracy, and Sheilagh Ogilvie. 2014. "Does the European Marriage Pattern Explain Economic Growth?" *The Journal of Economic History 74 (3): 651–93.*

Caveats

01

Sample bias

The inscriptions represent the middling-to-prosperous. The LHT signal refers to that stratum — which may have lagged behind the general population. Women in inscriptions are scarce commodity !

02

LLM role classification and age alignment

Ages and patronymics need careful manual review especially in long and complex inscriptions. Unnamed females (uxori, filiae carissimae) need to be relinked.

03

Christianisation

The C4 sample is tiny and features a commemorative shift. The C4 rebound after the trough is the weakest claim in this presentation.

The graveyard reopened

While Shaw and Saller made tombstones speak in the aggregate, ***computational temporal analysis lets them speak across time.***

Marriage age among women in ancient Rome was not a constant. It moved with the empire — rising in prosperity, falling under pressure, whether epidemic or economic stress and maybe even plain anxiety — almost as life history theory predicts.

Age of anxiety?

MATERIALISTIC VS IDEALISTIC EXPLANATIONS

Contra-Baumard

- marriage age drops despite affluent present before the exogenous shocks, and therefore not following LHT

Either:

- A) people somehow see the contraction on the horizon (LHT not anchored in environment, but in “perception”)
- B) spouses follow Wrigley & Schofield short-term high-wage/early-marriage trend of pre-industrial England

DANISH CASE

Comparanda

1800-1850: growth & transformation from absolutistic to democratic governance, Napoleonic wars, prosperity > rise in marriage age

1850 - 1900: economic well being, urbanisation, industrialisation BUT , catastrophic loss of Schleswig/Holstein leads to massive soul-searching and drop in marriage ages

Filtering as a methodological choice

We keep only inscriptions with both a coordinate pair and a chronological attribution.

What we sacrifice. Rav
better-preserved, mor

What we gain. The abi
undated inscription ca

```
# Patronymic filter: for daughters, keep only patronymic_review == "primary_role"
# to exclude filia-as-naming-convention cases (e.g. "Reducta Titi filia").
# Wives don't have patronymic issues, so the filter is a no-op for them.
women <- roles_joined %>%
  filter(role %in% c("wife", "daughter"),
         assigned_to == "deceased_role",
         gender == "F") %>%
  filter(role != "daughter" | patronymic_review == "primary_role") %>%
  mutate(
    # Prefer age_verified (from classifier), fall back to total_age (from EDH)
    age = coalesce(as.numeric(age_verified), total_age),
    group = case_when(
      category == "conjugal"   ~ "Wife (conjugal)",
      category == "descending" ~ "Daughter (descending)",
      TRUE ~ NA_character_
    )
  ) %>%
  filter(!is.na(group), !is.na(age), age > 0, age < 100)
```

SAMPLE COMPOSITION

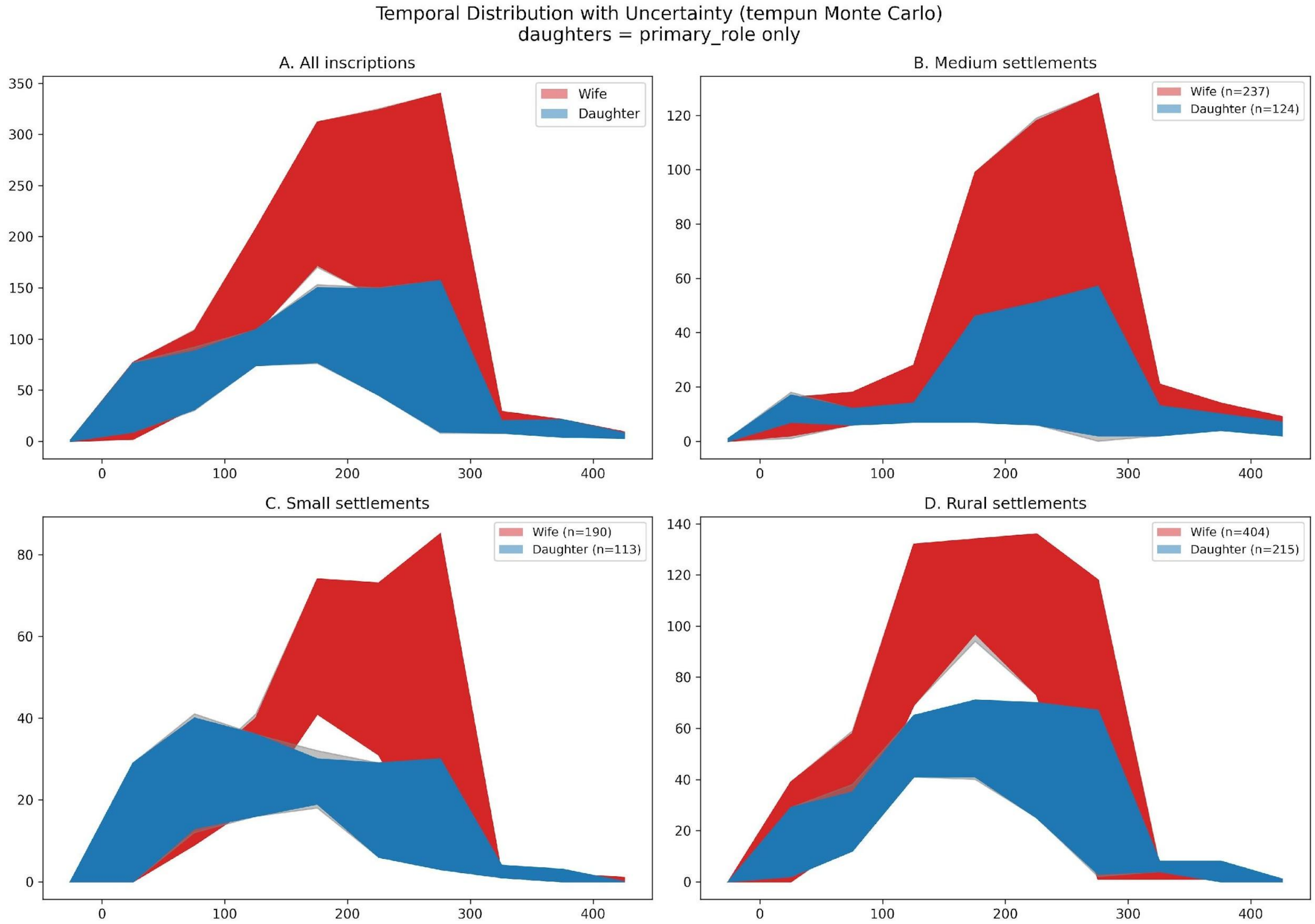
udy
IRE
required
only
audited
njugal

Urban contexts

Tempun. Each inscription's date range is sampled 200× from a uniform prior. We report the median crossover with 90% intervals across simulations.

What this shapes

Peak inscription density in C2; sharp dropoff after C3. Temporal resolution is best where the data are densest.



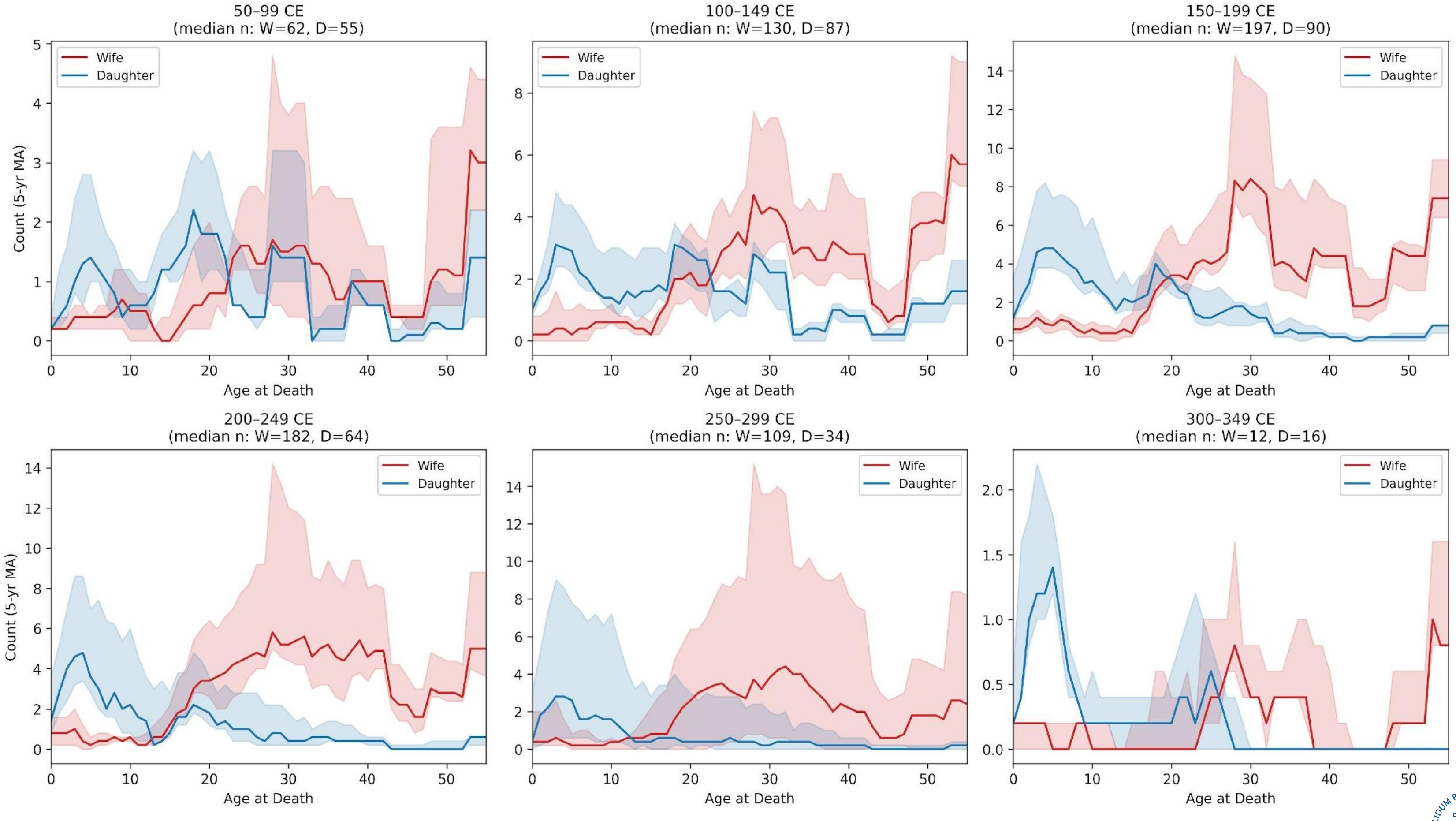
The same crossover ages in 50-year bins

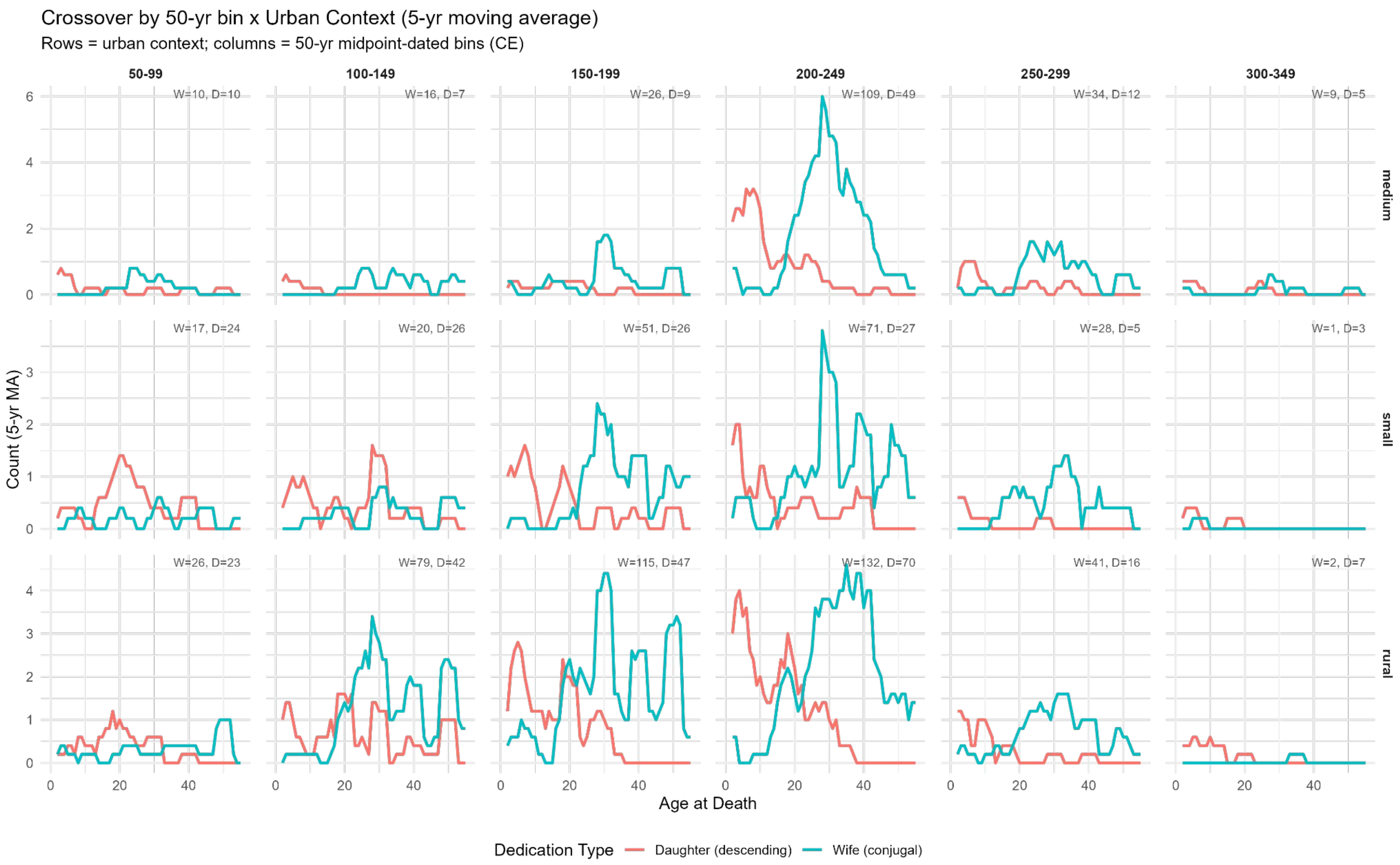
C1–C2: crossover clean-ish ~24 to 20.

200–249: compression toward younger ages (13-17!).

250–299: wide CIs, conjugal distribution destabilises.

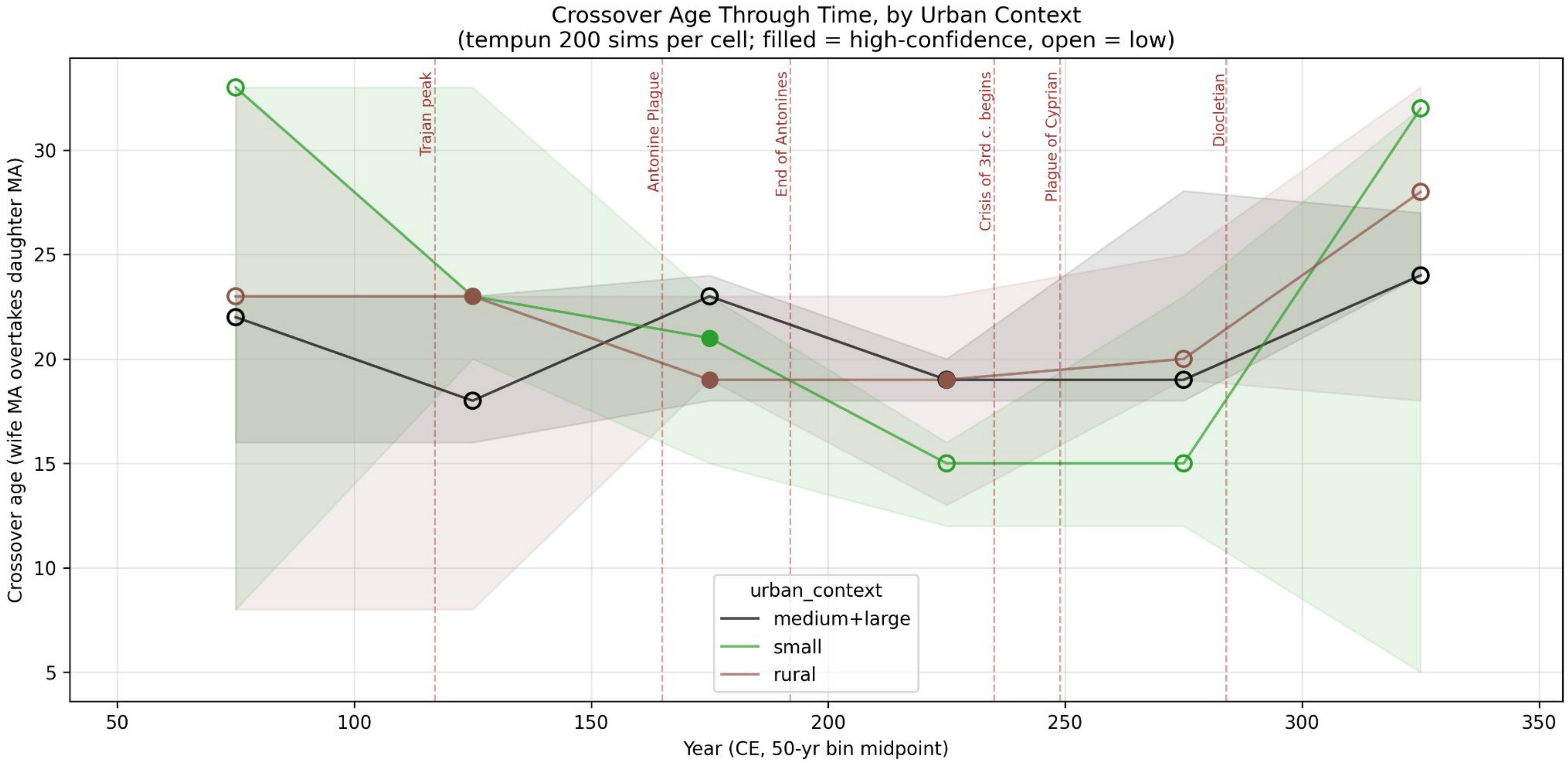
300–349: n too small to read.





Crossover by 50-yr bin x Urban Context (5-yr moving average)
Rows = urban context; columns = 50-yr midpoint-dated bins (CE)
50-99
100-149
150-199
200-249
250-299
300-349
medium
small
rural
W=10, D=10
W=16, D=7
W=26, D=9
W=109, D=49
W=34, D=12
W=9, D=5
W=17, D=24
W=20, D=26
W=51, D=26
W=71, D=27
W=28, D=5
W=1, D=3
W=26, D=23
W=79, D=42
W=115, D=47
W=132, D=70
W=41, D=16
W=2, D=7
Count (5-yr MA)
Age at Death
Dedication Type
Daughter (descending)
Wife (conjugal)

Crossover age through time - by urban context



Crossover age drops from ~23 in the late C1 to ~15 in the mid-C3 — the trough sits between the Antonine Plague and the Plague of Cyprian.

LLMs good at separating LOM from AAD (often miscoded)

Conjugal dedications to girls aged 5–15 — common in our raw extraction — turn out, on inspection, to be **duration phrases miscoded as ages at death**.

CORRECTLY ENCODED

VIXIT ANNIS XXII

“lived 22 years” ⟶ age at death = 22

MISCODED — DURATION READ AS AGE

VIXIT CVM VIRO ANNIS VII

“lived 7 years with her husband” ⟶ ***not*** *age 7*

Why this matters. Uncorrected, the bias pulls apparent marriage age systematically downward in the conjugal sub-population — exactly the group whose distribution drives the crossover. We identify, quantify, and correct these cases before any analysis.

Defining family roles with LLM

INSCRIPTION (CLEAN / INTERPRETIVE)

> *Servilla Praepusa filiae suae Lesbiae annorum XI mensium X hic sepultae sit levis terra quid sibi fata velint bellissima quae que creari edita laetitiae commoda si rapiuunt sed quae fatorum legi servare necesse est perverso lacrimas fundimus officio haec bis sex annos vix bene transierat ill lt a gt suas lacrimas nondum emiserat omnes et poterat semper flebilis esse suis parcite enim vobis tristes sine fine parentes parcius et manes sollicitare meos ponimus hunc titulum luctus solacia nostri qui legit ut dicat sit tibi terra levis*

HD042404 · id=HD042404 · person_id=1 · role=*daughter* · lemma=*filia* · case=nominative · total_age=**30** · age_match=match

MATCHED TEXT

Fuscia Proculi filia Secunda

INSCRIPTION (CLEAN / INTERPRETIVE)

> *Fuscia Proculi filia Secunda obita annorum XXX hic sita est heredes pudenti matri fecerunt*

DIPLOMATIC